
Statistical Inference

B. Statistical Data Science 2nd Year Indian Statistical Institute

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Exercise Series 9

Exercise 1. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}(-\theta, \theta)$, $\theta > 0$.

- (a) Check whether UMVUE of θ exists. If so, then identify the UMVUE.
- (b) Does the UMVUE attain the Cramér-Rao lower bound?

Exercise 2. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(\theta, 1)$, $\theta \in \mathbb{R}$.

- (a) Show that $T = \bar{X}^2 - 1/n$ is an unbiased estimator of θ^2 . Deduce that it is the UMVUE of θ^2 .
- (b) Find the variance of T .
- (c) Find the Cramér-Rao lower bound for the variance of an unbiased estimator of θ^2 in this model. Does T attain this lower bound?

Exercise 3. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(\mu, \sigma^2)$, where $\mu \in \mathbb{R}, \sigma^2 > 0$. Let $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ be the sample variance (with divisor $n-1$), which is an unbiased estimator of σ^2 . Recall that the MLE of σ^2 in this model is $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$.

- (a) Show that $\text{Var}(S^2) = \frac{\sigma^4}{n} \left(\kappa - \frac{n-3}{n-1} \right)$, where $\kappa = \mathbb{E}(X - \mu)^4 / \sigma^4$ is the kurtosis of X .

Note: For the normal distribution, $\kappa = 3$.

- (b) Find the MSE of $\hat{\sigma}^2$.
- (c) Verify that the MSE of $\hat{\sigma}^2$ is smaller than the MSE of S^2 .

Now, we consider a class of estimators of the form aS^2 , where $a > 0$ is a constant.

- (d) Show that $\text{MSE}(aS^2) = \mathbb{E}(aS^2 - \sigma^2)^2 = a^2 \text{Var}(S^2) + (a-1)^2 \sigma^4$.
- (e) Use parts (a) and (d) to establish that the estimator of the form aS^2 with the minimum MSE is $\frac{n-1}{n+1} S^2$.
- (f) Only assuming $X_1, \dots, X_n$ are i.i.d. with mean μ and variance σ^2 , show that $\text{MSE}(aS^2)$ is minimized at

$$a = \frac{n-1}{(n+1) + \frac{(\kappa-3)(n-1)}{n}}.$$

Note: aS^2 with the above choice of a is not an estimator, since a depends on the unknown parameter κ .

(g) Verify that

- (i) for distributions with $\kappa > 3$, the optimal a satisfies $a < \frac{n-1}{n+1}$.
- (ii) for distributions with $\kappa < 3$, the optimal a satisfies $\frac{n-1}{n+1} < a < 1$.

Exercise 4. An instrument is used to measure the radii of circles. However, the instrument takes the measurements with an additional $\text{Normal}(0, \sigma^2)$ noise. Suppose n independent measurements $R_1, \dots, R_n$ of circles with the same radius r (unknown) are made. Our parameter of interest is $g(r) = \pi r^2$, the area of the circle.

- (a) Find the MLE of $g(r)$. Is it unbiased?
- (b) If the estimator in part (a) is biased, then find an unbiased estimator of $g(r)$. Is it the UMVUE?
- (c) If the estimator in part (b) is not the UMVUE, find the UMVUE.

Exercise 5. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(\theta)$, $\theta \in [0, 1]$.

- (a) For $n \geq 2$, show that $X_1 X_2$ is an unbiased estimator of θ^2 .
- (b) Use part (a) to find the UMVUE of θ^2 .

Exercise 6. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Exponential}(\text{mean} = \theta)$, $\theta > 0$.

- (a) Find an unbiased estimator of θ based on $X_{(1)} = \min\{X_1, \dots, X_n\}$.
- (b) Find a better estimator than the one in part (a). Show that it is better.

Exercise 7. Let $X_1, \dots, X_{n+1} \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$, $p \in [0, 1]$. Define,

$$g(p) = \mathbb{P}_p \left(\sum_{i=1}^n X_i > X_{n+1} \right)$$

to be the probability that the sum of the first n observations exceed the $(n+1)$ -st.

- (a) Find a closed-form expression for $g(p)$.
- (b) Show that

$$T(X_1, \dots, X_{n+1}) = \begin{cases} 1 & \text{if } \sum_{i=1}^n X_i > X_{n+1}, \\ 0 & \text{otherwise,} \end{cases}$$

is an unbiased estimator of $g(p)$.

- (c) Use part (b) to find the UMVUE of $g(p)$.

Exercise 8. Let X be a single observation from a distribution with pdf

$$f_\theta(x) = \left(\frac{\theta}{2}\right)^{|x|} (1-\theta)^{1-|x|}, \quad x = -1, 0, 1, \quad 0 \leq \theta \leq 1.$$

- (a) Find the MLE of θ .

- (b) Define the estimator $T = T(X)$ by $T(x) = \begin{cases} 2 & \text{if } x = 1, \\ 0 & \text{otherwise.} \end{cases}$. Show that $\mathbb{E}_\theta(T) = \theta \forall \theta \in [0, 1]$.
- (c) Find a better estimator of θ than T . Show that it is better.

Exercise 9. Let X be a random variable with

$$\mathbb{P}_\theta(X = -1) = \theta, \mathbb{P}_\theta(X = x) = (1 - \theta)^2 \theta^x, x = 0, 1, 2, \dots$$

Here, $\theta > 0$ is an unknown parameter.

- (a) Show that $Z = Z(X)$ is an unbiased estimator of 0 if and only if $Z(x) = -xZ(-1) \quad \forall x = 0, 1, 2, \dots$
- (b) Use part (a) to show that $T = T(X)$ is an UMVUE if and only if $T(x) = T(-1) \quad \forall x = 1, 2, \dots$, $T(0)$ can be arbitrary.

Hint: T is UMVUE if and only if $\mathbb{E}_\theta(TZ) = 0$ for all unbiased estimators Z of 0.

- (c) Using part (b), verify that $g(\theta)$ has an UMVUE if and only if $g(\theta) = a + (b - a)(1 - \theta)^2$, for some $a, b \in \mathbb{R}$.

Hint: Every $g(\theta)$ must be of the form $\mathbb{E}_\theta(T)$ for some UMVUE T .

- (d) Find the UMVUE of $(1 - \theta)^2$.
- (e) Show that there exists an unbiased estimator of θ , but no UMVUE of θ exists.